WORKED SOLUTION

STEP I 2017, Question 12

Probability and Statistics • Expectation and Conditioning

STEP I 2017, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q12

The key idea

The organiser always collects the same takings, Nc; the only uncertainty is whether the prize has to be paid out at all. So the expected profit is the takings minus J times the probability that *somebody* picked the winning number, and everything comes down to the complementary event that *nobody* did. That probability is a power $(1-p)^N$, which is exactly what the approximation $(*)$ is built to handle.

Solution

Part (i)

Let W be the winning number and let E be the event that no participant picks W. Whatever W turns out to be, each participant independently avoids it with probability $1-\frac{1}{N}$, so

$$\mathrm{P}(E) = \left(1-\frac{1}{N}\right)^N.$$

The organiser receives Nc in every case. If E happens, nobody wins and the profit is Nc; otherwise the prize is shared among the winners, but the total paid out is J however many winners there are, so the profit is $Nc-J$. Hence

$$\text{expected profit} = Nc - J\,\mathrm{P}(E^c) = Nc - J\left[1-\left(1-\frac{1}{N}\right)^N\right].$$

Using $(*)$ with $a=1$, $(1-\frac{1}{N})^N \approx \mathrm{e}^{-1}$, so the expected profit is approximately $Nc - J(1-\mathrm{e}^{-1})$. If $J=2Nc$ this becomes

$$Nc - 2Nc\left(1-\mathrm{e}^{-1}\right) = Nc\left(\frac{2}{\mathrm{e}}-1\right).$$

Since $\mathrm{e} > 2$, we have $2/\mathrm{e} < 1$, so this is negative: the organiser expects to make a loss.

Part (ii)

There are γN popular numbers and $(1-\gamma)N$ unpopular ones. Each participant picks exactly one number, so the probabilities over all N numbers sum to 1:

$$\gamma N \cdot \frac{a}{N} + (1-\gamma)N \cdot \frac{b}{N} = 1 \quad \Longrightarrow \quad \gamma a + (1-\gamma)b = 1.$$

The organiser still chooses W uniformly, so W is popular with probability γ and unpopular with probability $1-\gamma$.

Ask yourself The participants are no longer choosing uniformly. Does that change how we find $\mathrm{P}(E)$?

Only through the probability of avoiding W, which now depends on which kind of number W is. So we condition on that. If W is popular, each participant independently avoids it with probability $1-\frac{a}{N}$; if unpopular, with probability $1-\frac{b}{N}$. By the law of total probability and $(*)$,

$$\mathrm{P}(E) = \gamma\left(1-\frac{a}{N}\right)^N + (1-\gamma)\left(1-\frac{b}{N}\right)^N \approx \gamma\,\mathrm{e}^{-a} + (1-\gamma)\,\mathrm{e}^{-b}.$$

As in part (i), the expected profit is $Nc - J(1-\mathrm{P}(E))$, so

$$\text{expected profit} \approx \gamma J\,\mathrm{e}^{-a} + (1-\gamma)J\,\mathrm{e}^{-b} + (Nc - J),$$

which is of the required form with

$$A = \gamma J, \qquad B = (1-\gamma)J, \qquad C = Nc - J.$$

The case $\gamma = \frac{1}{8}$, $a = 9b$. The relation gives $\frac{1}{8}\cdot 9b + \frac{7}{8}b = 1$, that is $2b = 1$, so

$$b = \tfrac{1}{2}, \qquad a = \tfrac{9}{2}.$$

With $J = 2Nc$ we have $A = \frac{1}{4}Nc$, $B = \frac{7}{4}Nc$ and $C = -Nc$, so the expected profit is approximately

$$Nc\left(\tfrac{1}{4}\mathrm{e}^{-9/2} + \tfrac{7}{4}\mathrm{e}^{-1/2} - 1\right).$$

The first term in the bracket is positive, so it is enough to show that $\frac{7}{4}\mathrm{e}^{-1/2} > 1$, that is $\sqrt{\mathrm{e}} < \frac{7}{4}$. Since $\mathrm{e} < 3$,

$$\sqrt{\mathrm{e}} < \sqrt{3} < \tfrac{7}{4},$$

where the last step holds because both sides are positive and $3 < \frac{49}{16}$ (that is, $48 < 49$). Hence the bracket is positive, and the organiser expects to make a profit. The prize is the same as in part (i), but because the winning number is often an unpopular one, it goes unclaimed more often.

Checks

- With $J = 2Nc$, part (i) gives an expected profit of about $(2/\mathrm{e} - 1)Nc \approx -0.264\,Nc$, while the final part gives $\left(\frac{1}{4}\mathrm{e}^{-4.5} + \frac{7}{4}\mathrm{e}^{-0.5} - 1\right)Nc \approx 0.064\,Nc$. A small profit, as claimed.
- For $N = 800$ the exact powers $(1 - a/N)^N$ give $0.0640\,Nc$, so the approximation $(*)$ is very good here.
- If $\gamma = 1$ (every number is "popular"), the relation forces $a = 1$ and the part (ii) formula reduces to part (i)'s $J\mathrm{e}^{-1} + Nc - J$.
- An exact enumeration by computer for $N = 4$ (all 4^4 ways the participants can choose) agrees with both unapproximated formulas.

Common slips

- Worrying about how the prize is shared. The organiser pays out J in total if there is at least one winner, whatever the number of winners.
- Assuming the winning number is popular with probability depending on a or b. The organiser picks uniformly, so it is popular with probability γ.
- Writing the relation as $a + b = 1$ or $\gamma a + b = 1$, forgetting to count how many numbers of each kind there are.

Beyond STEP

The number of participants who pick the winning number is binomial with parameters N and p, and $(*)$ is the first step in seeing that for large N and Np fixed it is approximately Poisson: $(1 - p)^N \to \mathrm{e}^{-Np}$ is just the Poisson probability of zero. Real lotteries with roll-overs are analysed in exactly this way, and players who choose unpopular numbers really do have a higher expected return. Grimmett and Welsh, *Probability: An Introduction*, develops the Poisson limit and conditioning arguments like the one used here.

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